Find the whole from a percentage

Use the scaffold, show each step and ask for an adult check when marked.

Name: ____ Date: ___________________

1. Read the model: 84 is 35% of a number. What is the number? Copy or complete the first step.

2. Find 25% of 360.

3. Find 35% of 480.

4. Miro says: Miro subtracts the percentage number rather than the percentage amount. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: 84 is 35% of a number. What is the number? Copy or complete the first step.
   Answer 240.

2. Find 25% of 360.
   Answer 90.

3. Find 35% of 480.
   Answer 168.

4. Miro says: Miro subtracts the percentage number rather than the percentage amount. Is this correct? Explain.
   Answer Calculate the percentage of the original amount first, then apply that change in the context.

Find the whole from a percentage

Work independently. Show a complete method and check at least one answer.

Name: ______________________________ Date: ____________

1. 84 is 35% of a number. What is the number?

2. Find 25% of 360.

3. Find 35% of 480.

4. Solve this independently and show every stage: 84 is 35% of a number. What is the number?

5. Use a second method or representation for: Find 25% of 360.

6. Check this result using an inverse, estimate or known fact: Find 35% of 480.

7. Prove or explain your reasoning: Find 17% of 650.

8. Identify and correct this misconception: Miro subtracts the percentage number rather than the percentage amount.

9. Write a complete sentence explaining how you checked one answer.

10. Write and solve a similar question to: Find 35% of 480.

Teacher answers and checking prompts

1. 84 is 35% of a number. What is the number?
Answer 240.

2. Find 25% of 360.
Answer 90.

3. Find 35% of 480.
Answer 168.

4. Solve this independently and show every stage: 84 is 35% of a number. What is the number?
Answer 240.

5. Use a second method or representation for: Find 25% of 360.
Answer Expected result: 90. Method or representation may vary.

6. Check this result using an inverse, estimate or known fact: Find 35% of 480.
Answer Expected result: 168. A valid check must be shown.

7. Prove or explain your reasoning: Find 17% of 650.
Answer 110.5. The explanation must show why the method works.

8. Identify and correct this misconception: Miro subtracts the percentage number rather than the percentage amount.
Answer Calculate the percentage of the original amount first, then apply that change in the context.

9. Write a complete sentence explaining how you checked one answer.
Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.

10. Write and solve a similar question to: Find 35% of 480.
Answer Answers vary. The new question must use the same mathematical structure and include a correct solution.

Find the whole from a percentage

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: ____ Date: __________________

1. Prove or explain your reasoning: Find 17% of 650.

2. Prove the answer to this question, rather than only calculating it: 84 is 35% of a number. What is the number?

3. Solve this in two different ways and compare them: Find 25% of 360.

4. Work backwards from the answer to reconstruct the question: 168.

5. Explain why this reasoning fails: Miro subtracts the percentage number rather than the percentage amount.

6. Create a new example that tests the same idea as: Find 35% of 480.

7. Find a second method or representation. Compare its efficiency with your first method.

8. Write one always, sometimes or never statement about today's learning and justify it.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

Teacher answers and checking prompts

1. Prove or explain your reasoning: Find 17% of 650.
   Answer 110.5. The explanation must show why the method works.

2. Prove the answer to this question, rather than only calculating it: 84 is 35% of a number. What is the number?
   Answer Expected result: 240. Proof or justification must match the lesson structure.

3. Solve this in two different ways and compare them: Find 25% of 360.
   Answer Expected result: 90. Two valid methods and a comparison are required.

4. Work backwards from the answer to reconstruct the question: 168.
   Answer One valid reconstruction is: Find 35% of 480. Other equivalent questions are acceptable.

5. Explain why this reasoning fails: Miro subtracts the percentage number rather than the percentage amount.
   Answer Calculate the percentage of the original amount first, then apply that change in the context.

6. Create a new example that tests the same idea as: Find 35% of 480.
   Answer Answers vary. The example and solution must preserve the mathematical structure.

7. Find a second method or representation. Compare its efficiency with your first method.
   Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.

8. Write one always, sometimes or never statement about today's learning and justify it.
   Answer Answers vary. A valid example and counterexample should support the classification.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.
   Answer Answers vary. The error must be mathematically relevant and the correction must be precise.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.
    Answer Answers vary. The explanation must distinguish the mathematical structure from the changed value.